```
here <- §(x,0)§[]
there <- §(x,10)§[]
gridx <- §10§[]
gridy <- §10§[]
resolution <- §16§[]
angle <- §NULL§[]
loc <- §NULL§[]
hhere <- §NULL§[]
max_angle <- §NULL§[]
step <- §NULL§[]
i <- §NULL§[]
```

```
BLOCK entry (angle, loc, hhere, there, resolution, gridx, max_angle, step, gridy, here, i)
[
  FROM LOOP AS entry
]
{
  gridx <- §{0} - 1§[gridx];
  gridy <- §{0} - 1§[gridy];
  step <- §(({0}.x - {1}.x) / {2}, ({0}.y - {1}.y) / {2}) :: point§[there, here, resolution];
  loc <- §{0}§[here];
  max_angle <- §NULL :: float§[];
  i <- §1§[];
  GOTO inter0 (angle, loc, there, resolution, gridx, max_angle, step, gridy, here, i)
}
```

```
BLOCK inter0 (angle, loc, there, resolution, gridx, max_angle, step, gridy, here, i)
[
  FROM entry
  WITH (angle, loc, there, resolution, gridx, max_angle, step, gridy, here, i)
  WHERE TRUE
]
{
  hhere <- §SELECT SUM(({0}! / ((c.x!) * (({0} - c.x)!))) * (u ** c.x) * ((1 - u) ** ({0} - c.x)) *
                     ({1}! / ((c.y!) * (({1} - c.y)!))) * (v ** c.y) * ((1 - v) ** ({1} - c.y))) AS h
          FROM   controlp AS c, LATERAL (SELECT {2}.x / {0}, {2}.y / {1}) AS _(u,v)§[gridx, gridy, here];
  JUMP loop0_head (angle, loc, hhere, there, resolution, gridx, max_angle, step, gridy, here, i)
}
```

```
BLOCK loop0_head (angle, loc, hhere, there, resolution, gridx, max_angle, step, gridy, here, i)
[
  FROM LOOP AS loop0_head
]
{
  condition%0 <- §{0} > {1}§[i, resolution];
  IF condition%0
  THEN GOTO truthy0 (angle, max_angle)
  ELSE GOTO falsey0 (loc, hhere, there, resolution, gridx, max_angle, step, gridy, here, i)
}
```

```
BLOCK truthy0 (angle, max_angle)
[
  FROM loop0_head
  WITH (angle, max_angle)
  WHERE (TRUE) AND (condition%0)
]
{
  EMIT §{0} = {1}§[angle, max_angle];
  STOP
}
```

```
BLOCK falsey0 (loc, hhere, there, resolution, gridx, max_angle, step, gridy, here, i)
[
  FROM loop0_head
  WITH (loc, hhere, there, resolution, gridx, max_angle, step, gridy, here, i)
  WHERE (TRUE) AND (NOT (condition%0))
]
{
  loc <- §({0}.x + {1}.x, {0}.y + {1}.y) :: point§[loc, step];
  GOTO inter8 (loc, hhere, there, resolution, gridx, max_angle, step, gridy, here, i)
}
```

```
BLOCK inter8 (loc, hhere, there, resolution, gridx, max_angle, step, gridy, here, i)
[
  FROM falsey0
  WITH (loc, hhere, there, resolution, gridx, max_angle, step, gridy, here, i)
  WHERE TRUE
]
{
  hloc <- §SELECT SUM(({0}! / ((c.x!) * (({0} - c.x)!))) * (u ** c.x) * ((1 - u) ** ({0} - c.x)) *
                    ({1}! / ((c.y!) * (({1} - c.y)!))) * (v ** c.y) * ((1 - v) ** ({1} - c.y))) AS h
          FROM   controlp AS c, LATERAL (SELECT {2}.x / {0}, {2}.y / {1}) AS _(u,v)§[gridx, gridy, loc];
  GOTO inter9 (hhere, loc, there, resolution, gridx, max_angle, step, gridy, here, i, hloc)
}
```

```
BLOCK inter9 (hhere, loc, there, resolution, gridx, max_angle, step, gridy, here, i, hloc)
[
  FROM inter8
  WITH (hhere, loc, there, resolution, gridx, max_angle, step, gridy, here, i, hloc)
  WHERE TRUE
]
{
  angle <- §degrees(atan(({0} - {1}) / sqrt(({2}.x - {1}.x)** 2 + ({2}.y - {1}.y)** 2)))§[hloc, hhere, loc];
  GOTO inter10 (angle, loc, hhere, there, resolution, gridx, max_angle, step, gridy, here, i)
}
```

```
BLOCK inter10 (angle, loc, hhere, there, resolution, gridx, max_angle, step, gridy, here, i)
[
  FROM inter9
  WITH (angle, loc, hhere, there, resolution, gridx, max_angle, step, gridy, here, i)
  WHERE TRUE
]
{
  condition%1 <- §{0} IS NULL OR {1} > {2}§[max_angle, angle, max_angle];
  IF condition%1
  THEN GOTO truthy1 (angle, loc, hhere, there, resolution, gridx, step, gridy, here, i)
  ELSE JUMP loop0_head (angle, loc, hhere, there, resolution, gridx, max_angle, step, gridy, here, i)
}
```

```
BLOCK truthy1 (angle, loc, hhere, there, resolution, gridx, step, gridy, here, i)
[
  FROM inter10
  WITH (angle, loc, hhere, there, resolution, gridx, step, gridy, here, i)
  WHERE (TRUE) AND (condition%1)
]
{
  max_angle <- §{0}§[angle];
  JUMP loop0_head (angle, loc, hhere, there, resolution, gridx, max_angle, step, gridy, here, i)
}
```